

ZELL AM SEE, Austria

WMO: 111440

Lat: 47.3267N

Lon: 12.7953E

Elev: 2526

StdP: 13.40

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	6.9	11.0	3.2	7.1	8.4	7.3	8.8	12.8	13.1	33.3	10.9	33.3	1.7	40	0.524

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.0	84.7	65.5	81.0	64.0	77.6	62.7	67.6	80.3	65.8	77.2	64.3	74.4	5.8	20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.2	95.3	71.5	62.0	91.1	69.9	60.6	86.9	68.3	33.4	80.4	32.0	77.5	30.8	74.8	73.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.7	11.3	9.4	DB	0.8	90.5	6.9	3.3	-4.2	92.9	-8.2	94.8	-12.1	96.6	-17.1	99.0	(4)
(5)			WB	-0.2	70.7	7.5	1.9	-5.6	72.1	-10.0	73.2	-14.2	74.3	-19.6	75.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	46.8	25.8	30.2	39.0	47.6	56.0	61.6	64.7	64.1	56.6	48.6	37.4	28.7	(6)	
	DBStd	15.26	7.80	7.43	7.10	7.31	6.55	7.16	6.38	5.93	6.21	6.65	7.12	7.02	(7)	
	HDD50	2970	751	554	345	127	22	6	0	2	15	106	381	661	(8)	
	HDD65	6839	1216	974	807	521	285	146	85	87	256	509	827	1126	(9)	
	CDD50	1798	0	0	3	56	207	353	457	440	215	63	4	0	(10)	
	CDD65	192	0	0	0	0	6	44	76	61	5	0	0	0	(11)	
	CDH74	1600	0	0	0	7	108	406	587	446	45	1	0	0	(12)	
	CDH80	410	0	0	0	1	16	107	170	113	3	0	0	0	(13)	
(14) Wind	WSAvg	3.5	2.5	2.8	3.7	4.2	4.3	4.2	4.1	3.8	3.5	3.2	2.9	2.5	(14)	
(15) Precipitation	PrecAvg	55.9	3.0	3.0	3.9	3.5	5.4	6.9	7.6	7.2	5.2	3.9	3.5	3.2	(15)	
	PrecMax	63.1	7.3	7.9	8.9	6.0	7.9	10.7	11.5	11.3	9.6	6.8	7.2	9.1	(16)	
	PrecMin	46.5	0.3	0.7	1.5	0.6	1.3	4.1	3.4	4.6	2.1	0.9	0.2	0.8	(17)	
	PrecStd	4.4	1.8	1.8	2.0	1.6	1.7	1.7	2.0	1.7	1.8	1.6	1.6	1.7	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.5	51.4	64.6	73.7	82.5	88.1	89.5	87.9	78.9	69.9	62.7	46.6	(19)	
		MCWB	42.1	41.9	49.2	54.9	61.5	65.9	68.5	67.4	63.6	57.0	50.3	41.9	(20)	
	2%	DB	39.5	45.0	59.5	69.3	77.0	83.7	84.6	83.2	74.0	64.9	54.5	41.4	(21)	
		MCWB	37.0	38.7	46.8	53.2	59.1	64.4	65.8	65.7	60.9	55.5	47.2	38.9	(22)	
	5%	DB	36.8	41.4	54.6	64.8	72.9	79.1	80.7	79.0	70.2	61.2	50.0	38.8	(23)	
		MCWB	35.2	36.7	44.6	50.6	57.2	62.9	64.5	64.3	59.2	53.6	45.9	37.0	(24)	
	10%	DB	35.0	38.9	49.8	60.8	68.7	74.8	76.6	75.1	66.6	57.8	46.6	36.6	(25)	
		MCWB	33.9	35.4	42.6	49.0	55.5	61.1	63.1	62.7	57.8	52.2	44.0	35.2	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.0	42.4	50.5	56.7	63.4	68.5	70.1	70.1	64.9	59.3	51.2	42.8	(27)	
		MCDB	45.5	49.0	61.8	70.7	76.5	83.0	84.9	84.1	75.5	66.5	59.0	45.5	(28)	
	2%	WB	37.4	39.5	47.6	53.9	60.5	66.1	67.7	67.4	62.6	56.7	48.3	39.5	(29)	
		MCDB	39.1	44.0	57.5	67.0	73.7	79.4	80.2	79.6	71.5	62.8	53.1	41.1	(30)	
	5%	WB	35.4	37.4	45.4	51.7	58.5	64.2	65.9	65.5	60.6	54.8	46.2	37.1	(31)	
		MCDB	36.3	40.3	53.2	63.0	69.7	75.5	76.6	76.2	68.0	59.9	49.8	38.3	(32)	
	10%	WB	33.9	35.9	43.1	49.7	56.6	62.4	64.3	63.7	58.8	53.0	43.9	35.4	(33)	
		MCDB	34.8	38.0	48.7	58.5	66.2	72.5	74.3	72.5	64.8	57.3	46.6	36.4	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	11.2	14.2	17.2	19.3	19.8	18.9	19.0	18.7	17.2	16.3	11.3	9.5	(35)	
		MCDBR	11.4	17.8	26.3	27.4	28.1	27.6	26.6	26.2	23.7	21.3	17.8	10.8	(36)	
	5% WB	MCWBR	8.8	12.1	14.3	12.9	12.0	10.4	10.4	10.8	11.6	12.2	11.3	8.2	(37)	
		MCWBR	10.3	15.3	24.3	25.3	25.2	24.9	23.8	24.0	21.4	19.7	16.1	10.0	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.276	0.306	0.351	0.394	0.409	0.416	0.415	0.407	0.370	0.340	0.308	0.277	(40)	
		taud	2.372	2.284	2.233	2.216	2.239	2.265	2.271	2.312	2.390	2.445	2.465	2.432	(41)	
	Ebn at Noon	Ebn at Noon	259	271	273	272	272	270	268	265	266	257	245	245	(42)	
		Edh at Noon	22	30	37	42	43	42	41	38	32	26	21	19	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	418	712	1027	1305	1494	1655	1684	1497	1098	764	431	342	(44)	
	RadStd	RadStd	50	90	122	165	180	169	159	176	144	82	73	53	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+1.10	N/A	N/A	+1.91	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (91 neighbors)		+0.77	N/A	N/A	N/A	+0.72	+0.72	N/A	N/A	+124	+59

Nomenclature: See separate page